Glasgow College, UESTC

Interim Report on UESTC4006P(BEng) Final Year Project

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Degree programme	Bachelor's Degree Programme
Academic year	4

Placement Company (if appropriate)	
Working Title of Project	Design and Implementation of Road Crack Detection System Based on YOLO Network Model
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Declaration of Originality and Submission Information	*I affirm that this submission is all my own work in accordance with the University of Glasgow Regulations and the School of Engineering requirements* Signed (Student): Bingyi Liu

Your report should be NO more than 12 pages in length and include the below subject headings and incorporated within this document:

1. **Project Outline (~1 page)**
2. **Work done so far [Methods, Models, Theory, Simulations, etc.] (at least 4 pages)**
3. **Challenges and Solutions (at least 1 page)**
4. **Conclusions from initial work and Future Work (at least 1 page)**
5. **Revised Gantt Chart**
6. **References (Not counted in pages)**

Interim (Mid-Project) Report

1. Project Outline

With the continuous expansion of road networks, automated and standardised detection of road defects has become a critical requirement for road maintenance and safety management. Road cracks, as the most visible manifestation of early pavement deterioration, significantly impact pavement durability, traffic safety, and subsequent maintenance costs. Traditional manual inspection methods are inefficient and highly subjective, struggling to meet engineering requirements for consistency, traceability, and real-time performance under complex lighting conditions, background texture interference, and large-scale road scenarios. Consequently, this project addresses practical road maintenance needs by developing a 'YOLO-based Road Crack Detection System Design and Implementation'. This system enables rapid localisation, precise segmentation, and output of crack detection results within road imagery/video, providing a foundation for subsequent assessment and decision-making.

This project employs the YOLOv8 and YOLO11 series as technical baselines, constructing a 'detection-segmentation' cascaded system rather than training two independent models. The specific workflow is as follows: first, the detection model (YOLOv8n) rapidly locates candidate regions of interest (ROIs) for cracks across the entire image. Subsequently, the image is cropped/scaled based on the detection bounding boxes, utilising sliding window segmentation and overlapping stitching where necessary to address issues of oversized ROIs and scale collapse. The ROI is then fed into the segmentation model (YOLOv8n-seg) for pixel-level segmentation. scaling based on detection bounding boxes, employing sliding window segmentation and overlapping stitching where necessary to address oversized ROIs and scale collapse issues. These ROIs are then fed into the segmentation model (YOLOv8n-seg) for pixel-level semantic segmentation. Finally, the segmentation results are overlaid onto the original image to generate visualised overlay maps, binary masks, and exportable structured outputs. The core advantage of this cascading strategy lies in: controlling computational costs while enabling the segmentation model to operate at scales and fields of view closer to its training distribution. This enhances its ability to represent elongated cracks, reduces computational pressure from full-image segmentation, and aligns more closely with the practical road inspection workflow of 'initial coarse-grained localisation followed by fine-grained quantification'.

Overall, this project has completed the baseline detection-segmentation cascade system and end-to-end workflow for the YOLOv8n series ahead of the mid-term milestone. Subsequent work will progressively enhance model accuracy and generalisation capabilities through custom evaluation datasets, comparative experiments, and targeted optimisations. The ultimate goal is to deliver video detection capabilities and a GUI-based solution, establishing an engineering-ready road crack detection system.

All source code for this final-year project is available on GitHub: https://github.com/ScatteredWood/UESTC4006P-Individual-Project.

2. Work Done So Far

2.1. Theory

Road crack detection is challenging primarily due to cracks' slender, low-contrast morphology and strong interference from shadows, lane markings, and pavement textures. In engineering inspection, outputs must further be verifiable and quantifiable, rather than merely "visible". Therefore, this project adopts a joint formulation that combines Object Detection for fast localisation and Semantic/Instance Segmentation for pixel-accurate delineation.

Within computer vision, detection and segmentation differ in both mathematical definition and optimisation objective. Following the YOLO paradigm proposed by Redmon et al. [1], object detection can be formulated as joint regression and classification: for a given road image, the detection head predicts bounding-box geometry and confidence scores as:

$$B = \{b_x, b_y, b_w, b_h, P_{obj}, C_{cls}\} \tag{1}$$

where the predicted terms describe the box centre offsets, normalised width/height, and confidence/class probability. Training minimises localisation regression (typically IoU-based) and classification losses to match ground truth. However, a rectangular box is inherently a coarse approximation for non-rigid crack shapes: it inevitably includes irrelevant background and cannot preserve the curved topology needed for subsequent quantitative measurement.

By contrast, semantic/instance segmentation refines the task into pixel-level mask generation. In efficient single-stage segmentation (e.g., YOLOv8-Seg), a common formulation follows the YOLACT idea proposed by Bolya et al. [2], which predicts shared prototype masks and instance-specific mask coefficients. For the k-th detected crack instance, the final binary mask is obtained by:

$$M_k = \sigma(P \cdot C_k^T) \tag{2}$$

where the network outputs prototype feature maps and combines them with instance coefficients followed by a Sigmoid activation. This mask-based representation enables accurate reconstruction of tortuous crack paths and boundaries. As highlighted in Mask R-CNN [3], pixel-level masks are also the prerequisite for deriving physical descriptors (e.g., length, average width, and area), which are not reliably obtainable from bounding boxes alone.

(a)

(b)

Figure 1 Comparison of output results for object detection (a) and semantic segmentation (b). (a) demonstrates that bounding boxes can only localise crack regions, encompassing substantial background areas; (b) demonstrates that segmentation masks can precisely

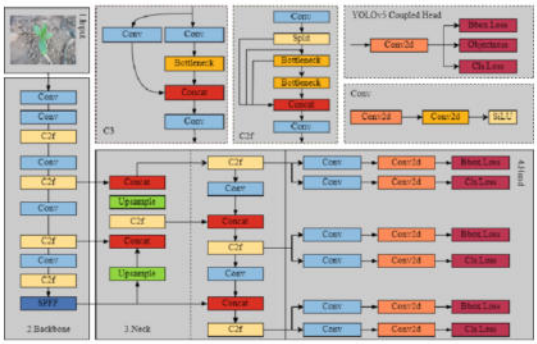


Figure 2 The segmentation branch principle based on prototype masks and coefficient prediction [2]. The model generates the final crack mask through a linear combination of prototype bases and instance coefficients, achieving efficient pixel-level segmentation

To balance speed and accuracy, this project adopts a detection-first cascaded pipeline. The detector performs high-throughput global screening and proposes ROIs, reducing the search space and suppressing global background clutter. The segmenter then conducts fine-grained delineation within ROIs. During cascading, dynamic cropping and resampling are applied so that cracks are processed at a scale closer to the segmentation model's training distribution. This decomposition of "global segmentation" into "localisation + local refinement" reduces computational burden while simultaneously outputting both location coordinates and pixel-level evidence [4].

2.2. Dataset Selection

To comprehensively evaluate the robustness and accuracy of road crack detection and segmentation algorithms in complex scenarios, this project selected two highly representative public datasets: RDD2022 and CRACK500. These were employed for training and evaluation in object detection and semantic segmentation tasks respectively.

2.2.1. RDD2022 Dataset (China_Drone & China_MotorBike)

RDD2022 (Road Damage Detection 2022) is a large-scale and diverse road damage dataset [5]. This project uses two China subsets—China_Drone and China_MotorBike—to obtain complementary viewpoints: the drone subset provides a top-down view covering broader road context and long-crack topology, while the motorbike

subset offers an ego-centric view that better reflects real inspection disturbances (e.g., illumination variation, shadows, and debris). Combining them forms a multi-perspective and multi-scale training set that enhances model robustness and generalisation.

Table 1 RDD2022 Dataset Label Information

Class ID	Label Name	Description	Explanation / Characteristics
0	D00	Longitudinal Crack	Cracks that run parallel to the direction of the road
1	D10	Transverse Crack	Cracks that run approximately perpendicular to the direction of the road
2	D20	Alligator Crack / Net Crack	Interconnected cracks forming a pattern resembling alligator skin or a net
3	D40	Pothole	A hole or depression in the road surface caused by material loss

2.2.2. CRACK500 Dataset

CRACK500 is a classic dataset in the field of pavement crack segmentation, comprising approximately 500 high-resolution pavement images (around 2000×1500 pixels) with corresponding pixel-level annotation masks. This dataset encompasses diverse crack morphologies on asphalt surfaces. Images, predominantly captured using mobile devices such as smartphones, exhibit inherent noise and blurring, closely mirroring real-world engineering image quality. Its meticulous pixel-level annotations provide high-quality supervisory signals for neural networks to learn crack edge details and slender structures.

2.3. Model Selection

YOLO (You Only Look Once), as a representative one-stage detector, models the detection problem as an end-to-end regression from image to bounding box and class probability, thereby enabling real-time inference [6]. Its modular trunk-neck-head architecture (feature extraction, multi-scale fusion such as FPN/PAN, prediction) proves particularly well-suited for fissure-like targets: at high resolutions, fissures may occupy only a few pixels and are susceptible to degradation through down sampling, necessitating robust multi-scale feature fusion and sufficiently fine-grained feature outputs [6][7].

YOLOv8 provides an integrated pipeline for training/validation/export and native multi-task support (detection and instance segmentation), offering a practical speed–accuracy trade-off and simplified deployment [8]. YOLOv11 extends the same Ultralytics framework with compatible pre-trained models and consistent inference/export interfaces, enabling controlled comparisons and ablation within one codebase [9]. Given the requirements for benchmarking YOLOv8 vs YOLOv11, exporting to ONNX/TensorRT, and GUI-oriented integration, adopting the Ultralytics YOLO family reduces engineering overhead and aligns with offline/real-time performance targets.

For segmentation, it is necessary to distinguish semantic segmentation (e.g., U-Net) from instance segmentation (e.g., Mask R-CNN) [10][11]. YOLOv8-seg follows the instance segmentation paradigm by adding

a mask branch to the detection head to predict per-instance masks [11][12][13]. As the target category in this project is limited to "crack", instance masks can be merged into a single-channel crack mask to support engineering-oriented post-processing (e.g., connected-component analysis, length/area statistics) and visual overlay [14][15].

Within the cascading strategy, the segmentation model is relieved of the search burden of 'finding cracks across the entire image'. Instead, it performs detailed contour delineation within detected ROIs (Regions of Interest). Consequently, it can focus more intently on pixel-level discrimination of slender structures [15][16]. This aligns with empirical findings in crack detection research, which indicate that segmentation of fine structures relies more heavily on local texture and boundary consistency [17].

2.4. Trains and Results

2.4.1. Detection

2.4.1.1. Experimental Setup

This section employs the RDD2022 dataset for experimentation, covering four categories of typical road defects (D00-D40) captured from both drone and motorcycle perspectives. The model undergoes transfer learning based on pre-trained YOLOv8n weights. To address the challenge of feature extraction for damage targets (e.g., elongated cracks), the input resolution was increased to 1024 pixels to enhance the model's representation capability for fine-grained textures. A composite augmentation strategy combining Mosaic and Rand Augment was introduced during training, alongside mixed-precision training (AMP) and an early-stopping mechanism. This approach aims to improve convergence efficiency while ensuring the model's generalisation performance.

Table 2 Training Configuration and Hyperparameters for Detection

Category	Parameter	Value / Setting
Model Architecture	Base Model	YOLOv8n (pre-trained)
	Input Resolution	1024×1024
Training Strategy	Epochs	300 (Patience=50)
	Batch Size	16
	Optimizer	Auto-selected (SGD/AdamW)
	Learning Rate	$lr_0 = 0.01$, Warmup=3
Loss Function	Weight Coefficients	Box: 7.5, Cls: 0.5, DFL: 1.5
Data Augmentation	Key Techniques	Mosaic, RandAugment, Erasing (0.4)
	Strategy Adjustment	Close Mosaic for the final 10 epochs

2.4.1.2. Evaluation Criteria

This section employs standard metrics from the object detection domain to quantitatively evaluate model performance.

1. **Intersection over Union (IoU)**

   IoU measures the overlap between a model-predicted bounding box B and a ground truth box $\hat{B}$, defined as follows:

$$IoU = \frac{Area(B \cap \hat{B})}{Area(B \cup \hat{B})} \tag{3}$$

2. **Precision and Recall**

   Based on predefined IoU and confidence thresholds, detection results are categorised as true positives (TP), false positives (FP), and false negatives (FN). Their respective formulas are:

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

3. **Average Precision (AP) and Mean Average Precision (mAP)**

   AP quantifies the area under the Precision-Recall curve, reflecting a model's detection performance on a single category. mAP is the arithmetic mean of AP across all N categories. The formulas are as follows:

$$AP = \int_0^1 p(r)dr \tag{6}$$

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i \tag{7}$$

Among these, the following two composite metrics warrant particular attention:

mAP@0.5: The mean AP across all classes when the IoU threshold is set to 0.5, primarily measuring the model's detection capability for targets.

mAP@0.5:0.95: The average mAP across a series of IoU thresholds ranging from 0.5 to 0.95 (with increments of 0.05). This metric imposes stricter demands on bounding box localisation accuracy, providing a more comprehensive reflection of the model's prediction quality.

2.4.1.3. Results Analysis

The training curves show stable optimisation: both training and validation losses decrease consistently and level off in later epochs, without noticeable oscillation. Precision/Recall and mAP@0.5 rise rapidly in the first 50–100 epochs and then plateau, while mAP@0.5:0.95 increases more gradually, indicating that coarse detection capability is learned early and fine localisation under strict IoU remains the main bottleneck. At convergence, the model achieves strong overall performance (Precision ≈ 0.88, Recall ≈ 0.84, mAP@0.5 ≈ 0.88, mAP@0.5:0.95 ≈ 0.56).

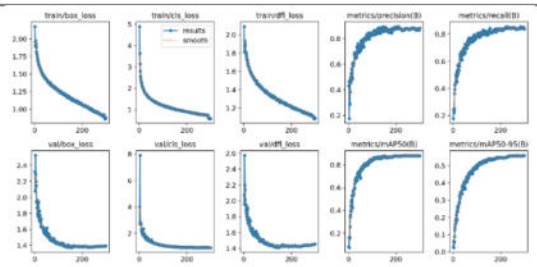


Figure 3 Training Curves for Detection

The normalised confusion matrix suggests good separability among the four defect categories (diagonal recall ≈ 0.89–0.90 with minimal cross-class confusion). The dominant errors occur at the disease–background boundary: around 0.08–0.10 true defects are missed as background (FN), and textured backgrounds are prone to being predicted as defects (background→D00/D10/D20/D40 ≈ 0.53/0.33/0.12/0.02).

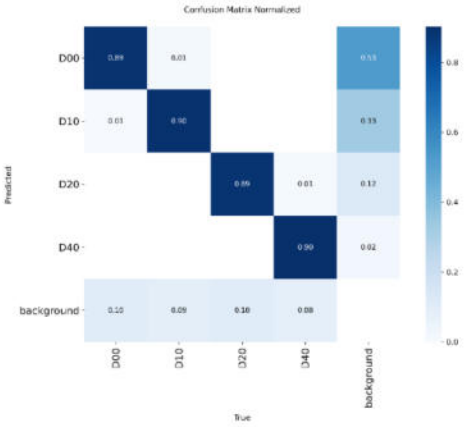
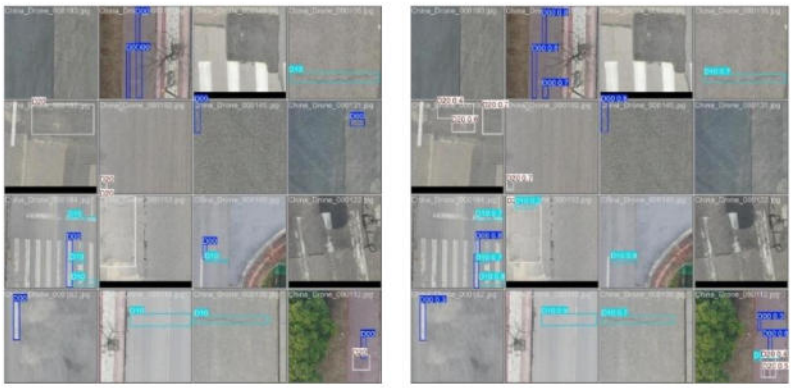


Figure 4 Normalized Confusion Matrix for Detection

Visual inspection further confirms reliable localisation for elongated cracks (notably D00/D10), but also reveals (i) duplicated or fragmented boxes within the same region and (ii) low-to-medium confidence false positives on complex textures, implying that thresholding/NMS and scale-aware post-processing remain key improvement directions.



(a) Label (b) Prediction

Figure 5 Visualization Examples for Detection

2.4.2. Segmentation

2.4.2.1. Experimental Setup

This section employs the lightweight YOLOv8n-seg model for pixel-level crack detection. The dataset adheres to the YOLO-Seg specification, simplifying the task to single-class instance segmentation. To balance computational efficiency with crack boundary detail, experiments were conducted at an input resolution of 800 pixels, utilising an Overlap Mask strategy with a down sampling ratio (mask ratio) set to 4. Training employs mixed-precision (AMP) and pre-trained weight initialisation, with 200 epochs and a 100-epoch early stopping threshold to prevent overfitting. For data augmentation, HSV colour dithering and geometric transformations (scaling, translation, flipping) were introduced to enhance generalisation capabilities. Mosaic augmentation was retained, while Mix-up was disabled to prevent texture artefacts in elongated crack structures.

Table 3 Training Configuration and Hyperparameters for Segmentation

Category	Parameter	Value / Setting
Task & Model	Architecture	YOLOv8n-seg (pre-trained)
	Task Type	Single-class Instance Segmentation
	Input Resolution	800×800
Training Setup	Epochs / Patience	200 / 100
	Batch Size	24
	Optimization	Auto (lr0=0.01, lrf=0.01)
	Regularization	Momentum=0.937, Weight Decay=5×10^{-4}
Mask Strategy	Mask Properties	Overlap Mask: True, Ratio: 4
Augmentation	Geometric	Translate (0.1), Scale (0.5), Flip-LR (0.5)
	Photometric	HSV (0.015, 0.7, 0.4)
	Composition	Mosaic (1.0), Mix-up (0.0)

2.4.2.2. Evaluation Criteria

Segmentation of cracks falls under the category of instance segmentation. By default, Ultralytics reports two sets of metrics simultaneously: bounding boxes (B) and segmentation masks (M). This dual reporting reflects the distinction between 'localisation capability' and 'pixel-level contour quality'. The core assessment is based on IoU (Intersection over Union): for both boxes and masks,

$$\mathrm{IoU}(B, B^{\mathrm{gt}}) = \frac{|\, B \cap B^{\mathrm{gt}} \,|}{|\, B \cup B^{\mathrm{gt}} \,|} \tag{8}$$

2.4.2.3. Results Analysis

The segmentation model exhibits steady convergence: train/seg_loss and train/box_loss decrease monotonically, while val/seg_loss drops sharply early and then enters a plateau with minor fluctuations,

suggesting that later epochs are dominated by generalisation error. Consistently, mAP improves rapidly in early training and then increases slowly towards stability, implying that subsequent training mainly trades off boundary refinement against suppressing FP/FN.

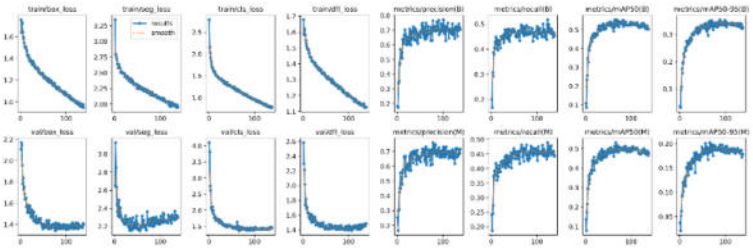


Figure 6 Training Curves for Segmentation

The confusion matrix indicates a clear recall limitation: only ≈ 0.53 of crack instances are correctly detected, while ≈ 0.47 are missed, reflecting a "high precision but low recall" conservative behaviour under current matching/threshold settings. Qualitatively, the model produces continuous masks for wide/high-contrast cracks, but struggles with narrow, low-contrast cracks and gravel-like high-frequency textures. Typical failures include local discontinuities or missing thin branches (recall loss), false positives on sharp edges/shadows (texture interference), and coarse/over-expanded boundaries that reduce performance under strict IoU (mAP@0.5:0.95). Overall, the model is usable for ROI-level crack localisation, but improving fine-structure representation and texture-artefact suppression is necessary for robust performance in complex backgrounds.

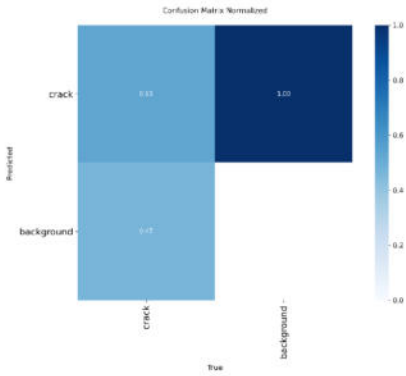
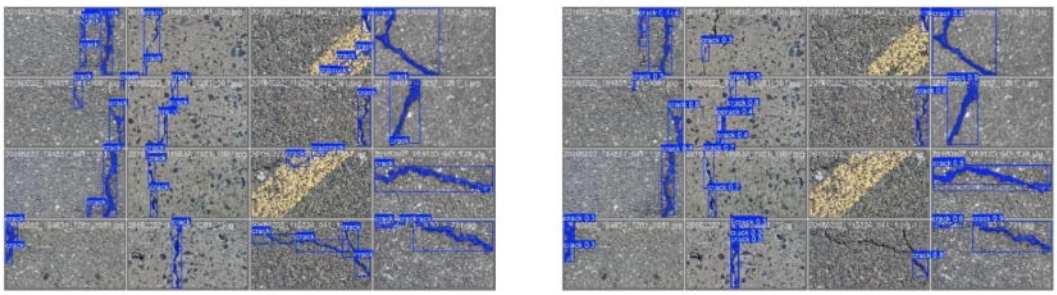


Figure 7 Normalized Confusion Matrix for Segmentation



(a) Label (b) Prediction

Figure 8 Visualization Examples for Segmentation

3. Challenges and Solutions

3.1. Introduction of ROI in Detection-Segmentation Cascade Architecture

In the task of road crack detection, the current approach employs a cascaded scheme of 'detection → segmentation'. The core function of the Region of Interest (ROI) module lies in transforming the raw input—characterised by 'global panoramic views, small targets, and a wide scale range'—into segmented network input that is 'localised close-up and scale-matched to the training distribution'. The specific workflow is as follows: First, an object detection model identifies crack candidate boxes (e.g., categories D00, D20) across the entire image. This stage prioritises high recall to minimise the risk of undetected defects. Subsequently, proportional padding is applied to the candidate boxes to prevent cropping edges from truncating crack textures. Next, semantic segmentation is performed on the pre-processed ROIs to generate local masks. Finally, these masks are remapped to their original coordinates on the global image, with overlapping regions merged via pixel-wise union operations. This achieves reconstruction from local fine contours to global defect distributions.

3.2. Current improvements

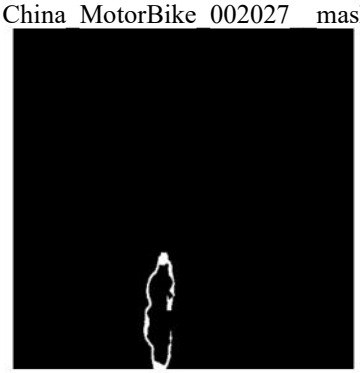
The baseline version (v2) exhibited two practical limitations: 'scale collapse' and 'texture interference'. When detection boxes cover excessively large areas, subsampling to scale ROIs to a fixed input scale often causes elongated crack features to become sub-pixel-level, leading to segmentation mask fractures or missed detections. Simultaneously, in high-frequency texture regions such as zebra stripes and patch seams, v2's fixed processing strategy struggles to distinguish structured boundaries from genuine cracks, readily generating false detections. Experimental comparisons indicate that v2 still requires improvement in coverage integrity for large-scale ROIs and robustness under strong textural backgrounds.

To address these shortcomings, this paper upgrades the ROI segmentation module in v3 to a comprehensive strategy combining 'explicit high-resolution + adaptive tiling + category-aware' approaches. Firstly, the resolution of segmentation inference is explicitly increased (e.g., to 1280 pixels), maximising the retention of fine crack geometry without altering weights. Secondly, an adaptive tiling mechanism is introduced for oversized ROIs. By performing segmented inference through sliding windows with overlapping regions followed by fusion, this effectively mitigates feature loss caused by global scaling. Additionally, v3 incorporates a class-aware parameter strategy, dynamically adjusting segmentation thresholds for different lesion categories. This is combined with CLAHE (Contrast-Level Auto-Levelling Enhancement) to enhance discrimination in complex backgrounds. Finally, the module's interpretability and parameter tuning efficiency are significantly improved through lightweight post-processing and multi-level visual debugging outputs.

(a) China_MotorBike_002027__overlay_v2

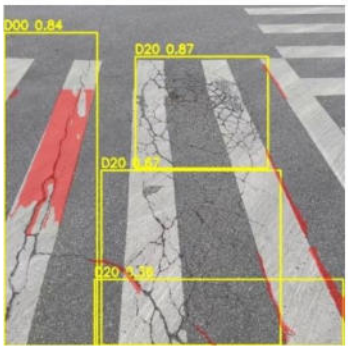
(b) China_MotorBike_002027__mask_v2

(c) China_MotorBike_002027__overlay_v3



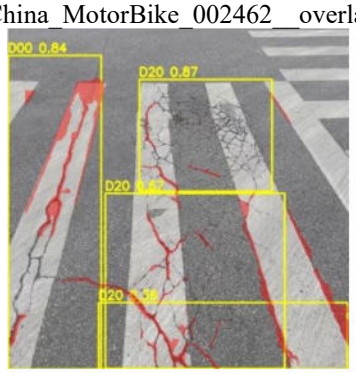
(d) China_MotorBike_002027__mask_v3

Figure 9 Scale Collapse



(a) China_MotorBike_002462__overlay_v2

(b) China_MotorBike_002462__mask_v2



(c) China_MotorBike_002462__overlay_v3

(d) China_MotorBike_002462__mask_v3

Figure 10 Texture Interference

4. Conclusions from initial work and Future Work

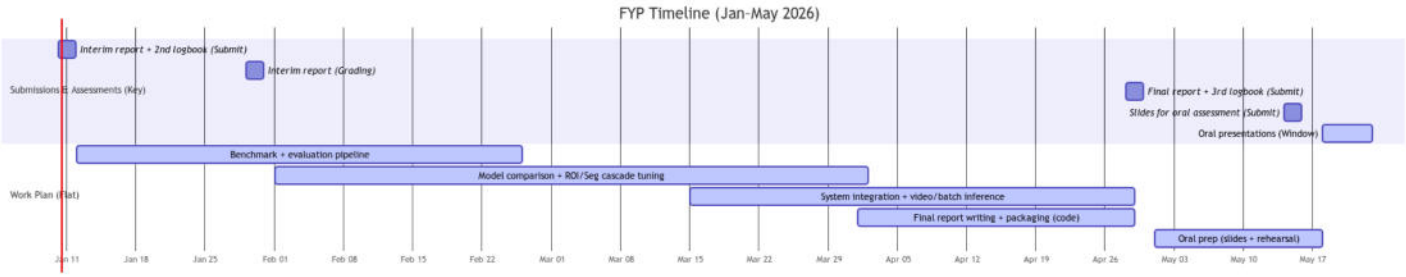
To date, the project has completed the following key tasks:

1. Mastered core principles of deep learning and YOLO-based object detection/segmentation, establishing the Ultralytics training and inference environment.
2. Trained detection models using RDD2022 (China_Drone and China_MotorBike subsets) as primary data sources, validating rapid candidate crack localisation in real-world road scenarios. Trained segmentation models using CRACK500 as the primary dataset, learning pixel-level features of crack's slender structures.
3. Implemented and validated a cascaded inference pipeline: Detection → ROI cropping/scaling → Segmentation → Result overlay output. This enables detection, segmentation, and visualised result output for image samples, laying foundations for subsequent expansion to batch processing and video inference.

In the next phase, the project will advance along the trajectory of 'benchmark construction → comparative optimisation → functional expansion → system integration' based on the existing cascaded baseline. Specific arrangements are as follows:

1. Establish a standardised evaluation benchmark: Create a high-quality local validation dataset encompassing diverse lighting conditions, multiple interference factors, and multi-material scenarios to provide a unified quantitative assessment standard for model iteration.
2. Compare multi-model performance: Contrast the accuracy (mAP) and speed (FPS) of YOLOv8n/s versus YOLOv11n/s, analyse architectural gains, and select the optimal baseline model balancing precision and efficiency.
3. Optimise models: Introduce lightweight attention mechanisms or multi-scale feature fusion modules to address missed detections of minute cracks, validating improvements through ablation studies.
4. Implement video stream detection: Extend single-frame detection workflows to continuous video streams, prioritising stability in temporal detection and offline processing efficiency.
5. System integration and GUI development: Package models and complete GUI design to deliver a road detection system supporting multi-source inputs, real-time visualisation, parameter configuration, and result export.

5. Revised Gantt Chart



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